

SC3103: Embedded Programming

Tutorial 1: FreeRTOS Concepts & Single-core Real-Time Scheduling

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1. Answer true or false with justification for the following FreeRTOS related questions.

- (a) When a user task must be blocked due to a semaphore, it changes state from “Running” to “Suspended”. **False, it goes to blocked, blocked = waiting, suspended = user blocked, only user can unblock**
- (b) ^{not scheduler}usStackDepth variable in the xTaskCreate() API represents the number of bytes allocated in stack for the task being created. **No. of words not no. of bytes**
- (c) XTaskGetTickCount() API returns the total number of ticks generated in the system since this task was created. **False, gives tick count from scheduler start**
- (d) Software timer callback functions are executed by the Timer Daemon task in FreeRTOS. **Timer daemon is basically a background task managing all software timers and call back functions, true**
- (e) If multiple readers are blocked on an empty Message Queue, the highest priority blocked reader will be unblocked when the queue is no longer empty (ties broken arbitrarily). **False**
- (f) If `configUSE_PREEMPTION = 0` and `configUSE_TIME_SLICING = 1`, then a non-preemptive (co-operative) scheduler with round-robin time slicing is deployed by FreeRTOS. **once non-preemptive, it cannot use round-robin thus false**

2. Consider the following sporadic real-time task set:

$$\tau_1 = (5, 2, 5), \tau_2 = (10, 3, 10), \tau_3 = (20, 5, 20).$$

Construct all possible **worst-case** schedules for this task set under Earliest Deadline First (EDF) scheduling algorithm.

3. Consider the following sporadic real-time task set:

$$\tau_1 = (10, 4, 7), \tau_2 = (5, 2, 7), \tau_3 = (10, 2, 10).$$

Construct any two possible **worst-case** schedules based on Deadline Monotonic (DM) and Least Laxity First (LLF) scheduling algorithms. Is this task set “schedulable” under DM/LLF?

Draw it out, then check the ties to make sure its okay, check until LCM of period if there are any missed deadline. if not, its schedulable. NOTE: EDF IS THE MOST OPTIMAL ON SINGLE CORE CPU ONLY, multi-core need to relook